

Hongdong Zhong

MIXED-SIGNAL IC DESIGNER

School of Integrated Circuits, Tsinghua University, Beijing, China

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"The ultimate ambition of technological advancement should be to return its benefits to humanity."

Research Interest

I am broadly interested in the intersection of **electronics**, **biology** and **medicine**, and how advances in these fields can improve human health and well-being. I am particularly interested in exploring novel circuit architectures, signal processing techniques, and system-level integration strategies to enhance the functionality and reliability of biomedical applications, such as implantable devices, wearable sensors, and neural interfaces. Additionally, I am keen on investigating the potential of emerging technologies, such as **Artificial Intelligence**, to revolutionize the way how interdisciplinary research is conducted and translated into practical solutions.

Education

School of Integrated Circuits, Tsinghua Univ.

M.S. IN INTEGRATED CIRCUITS SCIENCE AND ENGINEERING

Beijing, China

Aug. 2023 - Present

- Research Area: opAmp, VCO and PLL design, GPA: 3.91/4.00, Rank: 8/125

School of Integrated Circuits, Tsinghua Univ.

B.E. IN MICROELECTRONIC SCIENCE AND ENGINEERING

Beijing, China

Aug. 2018 - Jun. 2023

- Transferred from Basic Science Class (基科班), Dept. of Physics (Year 1-2) with a solid foundation in physics and mathematics
- Graduate with Distinction (Equivalent to Magna Cum Laude, Top 10%), GPA: 3.89/4.00, Rank: 3/42

Skills

EDA(Analog flow) Virtuoso, Spectre, AMS, HSPICE, ADS, EMX

EDA(Digital flow) Modelsim, VCS, Verdi, Xcelium, Design Compiler, ICC, Innovus

FPGA/Embedded Altium Designer, JLCEDA, Vivado, Arduino, MSP430, Raspberry Pi

Programming Python, C++, Shell, MATLAB, HTML, Verilog, VerilogA, System Verilog

Typesetting L^AT_EX, Markdown, Windows Word

Languages Mandarin(Native), Cantonese(Native), English(Fluent, Toefl: 106/120), Japanese(Basic), Spanish(Basic), Russian(Basic)

Others Obsidian, Feishu Base, Docker, OpenWrt, Qt, PyTorch

Publication

JOURNAL

A Time-Domain Accuracy-Boosted Temperature Compensated Crystal Oscillator

Junyang Yin, Wei Deng, Haikun Jia, Sining Pan, Minshuo Wang, Tianyu Chen, Hongdong Zhong, Ziyuan Guo, Zhihua Wang, Baoyong Chi

IEEE Journal of Solid-State Circuits (JSSC), Jul. 2025

Experience

Dept. of Electronic Engineering, Tsinghua Univ.

Beijing, China

TEACHING ASSISTANT OF FUNDAMENTALS OF ELECTRONIC CIRCUITS AND SYSTEMS (电子电路与系统基础)

Sept. 2025 - Present

- The only TA for a core undergraduate course *Fundamentals of Electronic Circuits and Systems* with 40+ students.
- Conducted Cadence Virtuoso tutorials which were recorded and uploaded on to public platform **Bilibili.com** (received 2,500+ views in total).
- Assisted professor in hybrid teaching setups, tutorial arrangements, and grading assignments, and conducting Q&A sessions to support student learning.

School of Integrated Circuits, Tsinghua Univ.

Beijing, China

PARTICIPANT

Mar. 2025 - Present

- In charge of the design of a low-phase-noise, wide-tuning-range voltage-controlled oscillator (VCO) engineered to ensure performance across a wide and dynamic thermal conditions.

School of Integrated Circuits, Tsinghua Univ.

Beijing, China

PARTICIPANT IN TWO TAPE-OUT PROJECTS

Mar. 2024 - Dec. 2024

- Taped out a Bandgap Reference chip using TSMC 180nm CMOS process, which achieved an integrated output noise of $<38 \mu\text{Vrms}$ across a wide supply voltage range of 1.6V to 3.6V.
- Taped out a modified RISC-V CPU core using TSMC 180nm CMOS process, which was based on the open-source project *tinyriscv*.

Warwick Summer School 2023

Coventry, England

PARTICIPANT IN INTRODUCTION TO ARTIFICIAL INTELLIGENCE

Jul. 2023 - Aug. 2023

- Collaborated closely with international peers to develop an AI agent program for an auction game during the 3-week study.
- Earned a first-class grade of A+ and Santander Scholarship.

BriRadio Microelectronics Technology Co., Ltd.

Beijing, China

MIXED-SIGNAL IC DESIGN INTERN

Sept. 2022 - Jun. 2024

Extracurricular Activities

Tsinghua University Science Museum

Beijing, China

TECHNICAL ADVISOR

Oct. 2024 - Mar. 2025

- Provided technical support during the curation phase of *The Era of Airwaves: An Exhibition on the Early History of Radio Technology*.

Beijing Organising Committee for the 2022 Olympic and Paralympic Winter Games

Beijing, China

RADIO FREQUENCY ASSISTANT (VOLUNTEER)

Dec. 2021 - Feb. 2022

- Assisted the technical team with electromagnetic environment monitoring in National Stadium (The Bird's Nest).
- Ensured compliance of wireless devices within allocated frequency bands and resolved sudden technical problems.
- Accumulated 200+ service hours, and was awarded the "Five-Star Zijing Volunteer" (五星级紫荆志愿者) title, the highest volunteer honor at Tsinghua University.

Honors & Awards

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| 2026 | Hong Kong PhD Fellowship (HKPFS) (2026) , Research Grants Council (RGC) |
| 2025 | Tsinghua Friends - Wuhan Elite Talent Scholarship (2025) , Tsinghua Univ. |
| 2023 | Graduate with Distinction (2023) , Tsinghua Univ. |
| 2022 | Tsinghua Friends - Suzhou Industrial Park Scholarship (2022) , Tsinghua Univ. |
| 2021 | Comprehensive Excellence Scholarship (2021) , Tsinghua Univ. |
| 2019 | Tsinghua Alumni - Yang Zhenbang Scholarship (2019) , Tsinghua Univ. |
| 2019 | Academic Excellence Scholarship (2019) , Tsinghua Univ. |

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